problems/MHD/fluxEmergenceSon2025DSGS

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. This case loads the physics of `fluxEmergenceSon2025` verbatim (`initialize.jl`, `user_flux.jl`, `user_source.jl` and `user_bc.jl` are one-line `include`s of the sibling's files) and has its own `user_inputs.jl` and `user_primitives.jl`. Its purpose is to keep the solution admissible *without any limiter*, through the reference-weighted form of the conserved DynSGS operator.

1 The problem as set up

Identical to `fluxEmergenceSon2025` (see that note: GLM-MHD, $\gamma = 1.05$, gravity, magnetostatic reference $q_e(z)$, well-balanced correction, absorbing layer, ψ damping $\alpha_p c_h/\Delta h$, periodic in x with walls), except for the deck values below.

Deck key / item	Value as run	Source / remark
Mesh	`FE_80x35.msh` refined once by p4est $\Rightarrow 160 \times 70$ quads of $0.5\,H_0$	`:linitial_refine => true`, `:init_refine_lvl => 1`
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 4$	
Time integration	`CarpenterKennedy2N54`, **no stage limiter**; fixed $\Delta t = 5 \times 10^{-3}\tau_0$, $t_{end} = 50\,\tau_0$	sibling: 7.5×10^{-3}, 54
Variables	`TOTAL()`, `"energy"`	
Sources	`:lsource => true` (as the sibling)	
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false	

2 DynSGS: deck keys and effective values

Same constants as the sibling; the one new key is `:dsgs_ref_weight`.

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`, `:lvisc => true`	
$:\mu$	all 1	
`:dsgs_sensor`	`"legacy"`	
`:dsgs_CR`, `:dsgs_Cmax`, `:dsgs_Cmin`	1.0, 0.5, 0.03	
`:dsgs_gamma`, `:dsgs_Prt`	$= \gamma_{mhd} = 1.05$, 1.0	
`:dsgs_nazarov_energy`	true	split energy flux
`:dsgs_norms`, `:dsgs_local_rel`	`"element"`, 1.0	
`:dsgs_conserved`	true	

Deck key / item	Value as run	Source / remark
`:dsgs_ref_weight`	**true**	the only case that sets it
`:dsgs_nodal_rho`	true in the deck, inactive	forced off by `:dsgs_conserved`
`:dsgs_fast_floors`, `:dsgs_thermal_residual`	false (defaults)	
`:dsgs_rel`, `:dsgs_hold_steps`	1.0, 2 (defaults)	
`:dsgs_cutoff`, `:dsgs_freeze_stage`, `:dsgs_reference`, `:ldsgs_nodal`	0, false, false, false	defaults

3 DynSGS as it runs for this case

$\Delta_K = \Delta_K^{elem}/(N+1) = 0.1\,H_0$. Steps 1–5 are those of the sibling, unchanged: rotation once per step, two held steps ($\nu = \nu_{\mathrm{fl}}$), the legacy residual against the assembled inviscid RHS (no wall-node zeroing on that path), element normalization with floors $r_{loc}(\bar\rho_K, \bar\rho_K \bar c_K, \bar\rho_K \bar c_K^2, \sqrt{\bar\rho_K}\bar c_K)$, max over the 8 non-$\psi$ slots, cap $C_{\max}\Delta_K \max(|\mathbf{v}| + c_f)$, and

$$\nu_c = \max(0, \min(0.5\,\Delta_K w_K,\ \Delta_K^2 \mathcal{R}_K)), \quad \nu_{\mathrm{fl}} = 0.03\,\Delta_K w_K, \quad \nu_K = \max(\nu_c, \nu_{\mathrm{fl}}), \quad \kappa_K^E = \max(0.0525\,\nu_c, \nu_{\mathrm{fl}}).$$

The residual sensor itself does *not* see the reference weighting: it acts only in the assembly.

4 The viscous operator: reference-weighted conserved form

`rhs.jl:1530` sets `dsgs_ref_weight[] = true` and `rhs.jl:1533 dsgs_split_energy[] = true`. `user_primitives.jl` returns, with $\rho_e = \rho_e(z)$ and $\delta q = q - q_e$,

$$\text{slots 1–5}: \ \frac{\delta\rho}{\rho_e},\ \frac{\delta(\rho u)}{\rho_e},\ \frac{\delta(\rho v)}{\rho_e},\ \frac{\delta E_{nth}}{\rho_e},\ \frac{\delta(\rho w)}{\rho_e}; \qquad \text{slots 6–9}: \ \delta B_x, \delta B_y, \delta B_z, \psi; \qquad \text{slot 10}: \ \rho_e; \quad \text{slot 11}: \ \frac{\delta E_{th}}{\rho_e},$$

and `_expansion_visc!` multiplies the coefficient of slots 1–5 by the weight ρ_e read at the quadrature point (`rhs.jl:2978`):

Equation	Viscous flux $\mathbf{F}_V$	
ρ	$\nu\rho_e\nabla(\delta\rho/\rho_e)$	mass diffusion on
$\rho u, \rho v, \rho w$	$\nu\rho_e\nabla(\delta(\rho u)/\rho_e), \ldots$	
E	$\nu\rho_e\nabla(\delta E_{nth}/\rho_e) + \kappa^E\rho_e\nabla(\delta E_{th}/\rho_e)$	weight on both parts
B_k, ψ	$\nu\nabla\delta B_k$, $\nu\nabla\psi$	unweighted

All in divergence form (conservative), zero at rest, no $\tau\cdot\mathbf{v}$ or resistive-work term. Where q_e is negligible against q it reduces to $\nu\nabla q$ on the fluid slots.

5 Positivity

- `:lpositivity` false, no stage limiter, no state modification: nothing clips, nothing is added. This is the point of the case.
- What remains: the flux-only floor $p \leftarrow \max(p, 10^{-9})$ in `user_flux!` (shared with the sibling; the state is not changed), the mass diffusion, the $C_{\min}$ background and the reference-weighted operator, which diffuses the *relative* departure ρ/ρ_e and so cannot carry a -10% departure of the dense side of the transition region across the $25\times$ jump in absolute terms (the mechanism the sibling needed its floor for, according to the deck header).

6 Observations: code versus comments

1. [**Fixed in `f03762b`.**] At `dcc3653` the deck header and the README said the case has "the SAME physics, mesh, initial and boundary conditions, time step and DynSGS constants" as the sibling; the mesh is refined once here (160×70 against 80×35) and Δt is 5×10^{-3} against 7.5×10^{-3} (t_{end} 50 against 54), values set in `7959928`. The comments on Δt (now: CFL ≈ 0.29 initially, ≈ 0.58 late on the refined mesh), on the $\tau \cdot u$ term and on the energy slot had the sibling's staleness. Header, comments and README now match the deck; the README's measured results are marked as obtained with $\Delta t = 7.5 \times 10^{-3}$ and not re-measured.

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	**Where (file:line)**
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report	`src/io/mod_inputs.jl:456-477`
`:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H) + 1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\,\mathbf{B} \cdot \nabla\mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs} + 2$ in 2D, $n_{eqs} + 1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)